

Cortex-M4 Exceptions and Interrupts

Ankit Srivastava

EE2028

ARMv7-M Architecture Ref Manual

DS12024Rev4 STM32L4Sxxx Datasheet

RM0432 STM32L4+ Technical reference manual

PM0214 STM32 Cortex-M4 Programming Manual

Figures from “The Definitive Guide to ARM® Cortex®-M3 and Cortex®-M4 Processors” by Joseph Yiu

Acknowledgement: Slides adapted from Dr Rajesh Panicker

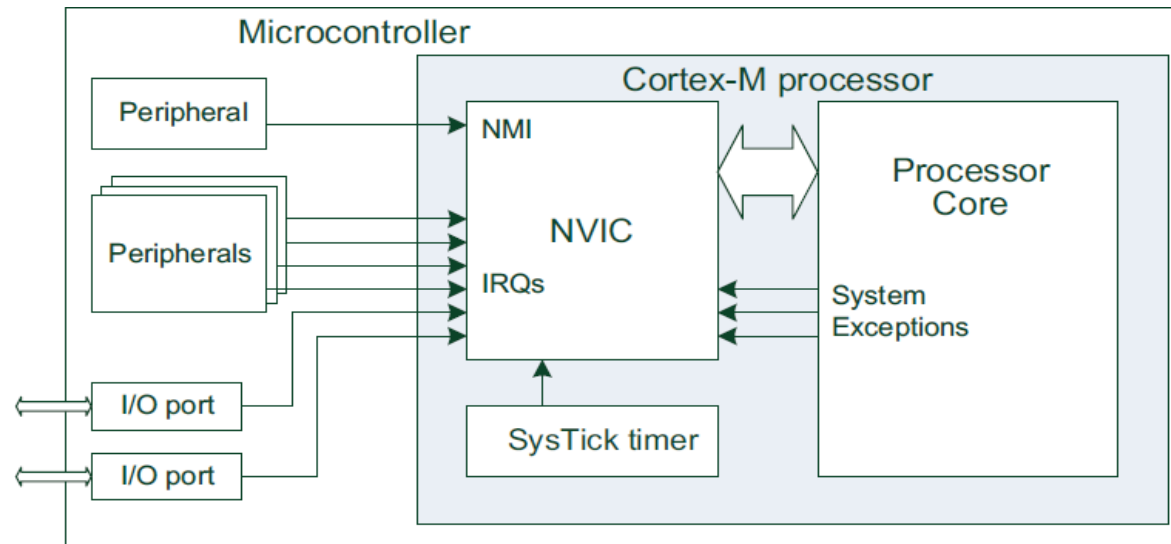


OVERVIEW

- Introduction to Cortex-M4 NVIC
- Cortex-M4 Interrupt system
- Vector Table
- NVIC Connections

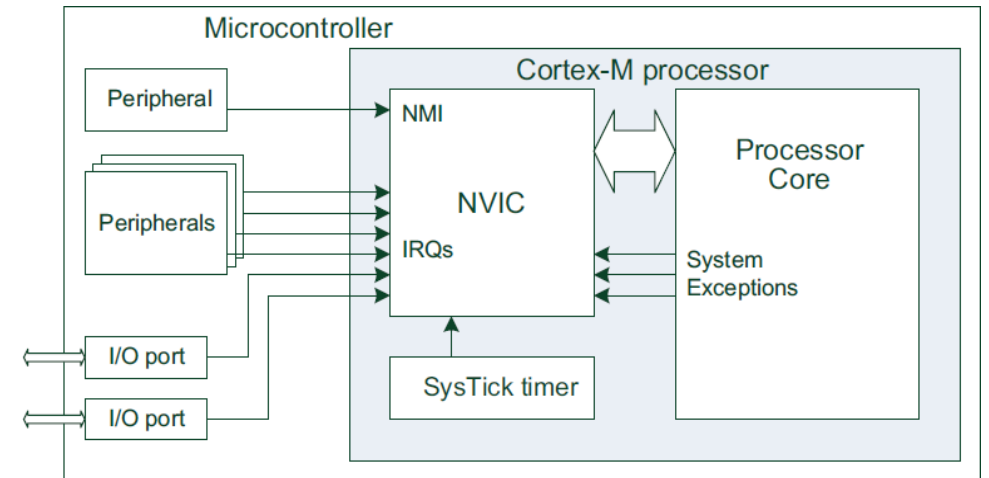
INTRODUCTION TO CORTEX-M4 NVIC

- Nested Vectored Interrupt Controller
 - is a 'core peripheral', closely coupled to the processor core
 - handles Exceptions and Interrupts with low latency
 - contains control logic and control registers for interrupt processing



INTRODUCTION TO CORTEX-M4 NVIC

- NVIC supports
 - up to 240 external interrupt inputs (IRQs) with programmable priority levels of 0-255
 - Exact number depends on the specific microcontroller / SoC
 - In the current STM32 Chip used in lab, the NVIC supports 95 vectored interrupts with 16 programmable levels (4 priority bits)
 - nesting of interrupts
 - dynamic reprioritization for interrupts
 - an external Non-Maskable Interrupt (NMI)
 - re-locatable vector table



CORTEX-M4 INTERRUPT SYSTEM

Cortex-M4 interrupts are classified into 2 groups

- System Exception ("Exception")
 - 15 for all Cortex M4 based chips
 - Reset, NMI, Hard fault, Bus fault, etc.
 - Generally generated within Cortex M4
 - E.g. SysTick
- External Interrupt ("Interrupt")
 - Up to 240, depending on the SoC
 - Generally generated by / taken in through on-chip peripherals
 - E.g. Push-button
- Every exception and external interrupt has an exception number (note: starts with 1), which is fixed for the interrupt/exception : 1-15 for system exceptions, 16 onwards for external interrupts
- A related number is IRQn, which is nothing but <exception_number> - 16 defined in stm32l4s5xx.h (Drivers>CMSIS>Device>ST>STM32L4xx>Include) and is used by NVIC driver functions which can be found in core_cm4.h
- (Drivers>CMSIS>Include)
 - IRQn is <0 for system exceptions and >=0 for external interrupts

CORTEX-M4 EXCEPTIONS AND INTERRUPTS

Exception Number	Exception Type	Priority	Descriptions
1	Reset	−3 (Highest)	Reset
2	NMI	−2	Non-Maskable Interrupt (NMI), can be generated from on chip peripherals or from external sources.
3	Hard Fault	−1	All fault conditions, if the corresponding fault handler is not enabled
4	MemManage Fault	Programmable	Memory management fault; MPU violation or program execution from address locations with XN (eXecute Never) memory attribute.
5	Bus Fault	Programmable	Bus error; usually occurs when AHB interface receives an error response from a bus slave (also called <i>prefetch abort</i> if it is an instruction fetch or <i>data abort</i> if it is a data access). Can also be caused by other illegal accesses.
6	Usage Fault	Programmable	Exceptions due to program error or trying to access co-processor (the Cortex-M3 and Cortex-M4 processor do not support co-processors).
7–10	Reserved	NA	–

7–10	Reserved	NA	–
11	SVC	Programmable	SuperVisor Call; usually used in OS environment to allow application tasks to access system services.
12	Debug Monitor	Programmable	Debug monitor; exception for debug events like breakpoints, watchpoints when software based debug solution is used.
13	Reserved	NA	–
14	PendSV	Programmable	Pendable service call; An exception usually used by an OS in processes like context switching.
15	SYSTICK	Programmable	System Tick Timer; Exception generates by a timer peripheral which is included in the processor. This can be used by an OS or can be used as a simple timer peripheral.

Exception Number	Exception Type	Priority	Descriptions
16	Interrupt #0	Programmable	It can be generated from on chip peripherals or from external sources.
17	Interrupt #1	Programmable	
...	...	...	
255	Interrupt #239	Programmable	

The term 'external' can be a bit confusing. Anything generated outside the Cortex M4 system is considered an 'external' interrupt by NVIC, even though they are generated by on-chip peripherals (e.g. : I2C2_ER interrupt). Interrupts which are truly 'external' to the chip (e.g. interrupts from sensors) are processed and passed to NVIC by EXTI block, which is an on-chip peripheral

VECTOR TABLE

- When an exc./int. takes place and is being handled by CPU, the CPU needs to locate the starting address of exc./int. handler
- Vector table is a part of memory used to store the starting addresses of the
- various exception/interrupt handlers
- Vector address is arranged according to exc./int. number $\times 4$
- Processor reads this table and loads the starting address of handler into PC
- Vector table is nothing but an array of function pointers pointing to interrupt handlers, with the exception number used as the array index

Table 7.6 Exception Vector Table After Power Up

Address	Exception Number	Value (Word Size)
0x00000000	—	MSP initial value
0x00000004	1	Reset vector (program counter initial value)
0x00000008	2	NMI handler starting address
0x0000000C	3	Hard fault handler starting address
...	...	Other handler starting address

VECTOR TABLE

- Vector table can be found in startup_stm32l4s5vitx.s (Core>Startup)
 - The file also provides weak (meant to be overridden by the user) aliases for each exception handler to the Default_Handler
 - The actual handlers can be found in stm32l4xx_it.h (Core>Inc)
- By default, vector table starts at memory address 0. Vector table can be relocated to another memory location by setting Vector Table Offset Register (VTOR) in SCB

Bits	Name	Type	Reset Value	Description
31:7	TBLOFF	R/W	0	Vector Table offset value

SCB->VTOR (0xE000ED08)

VECTOR TABLE

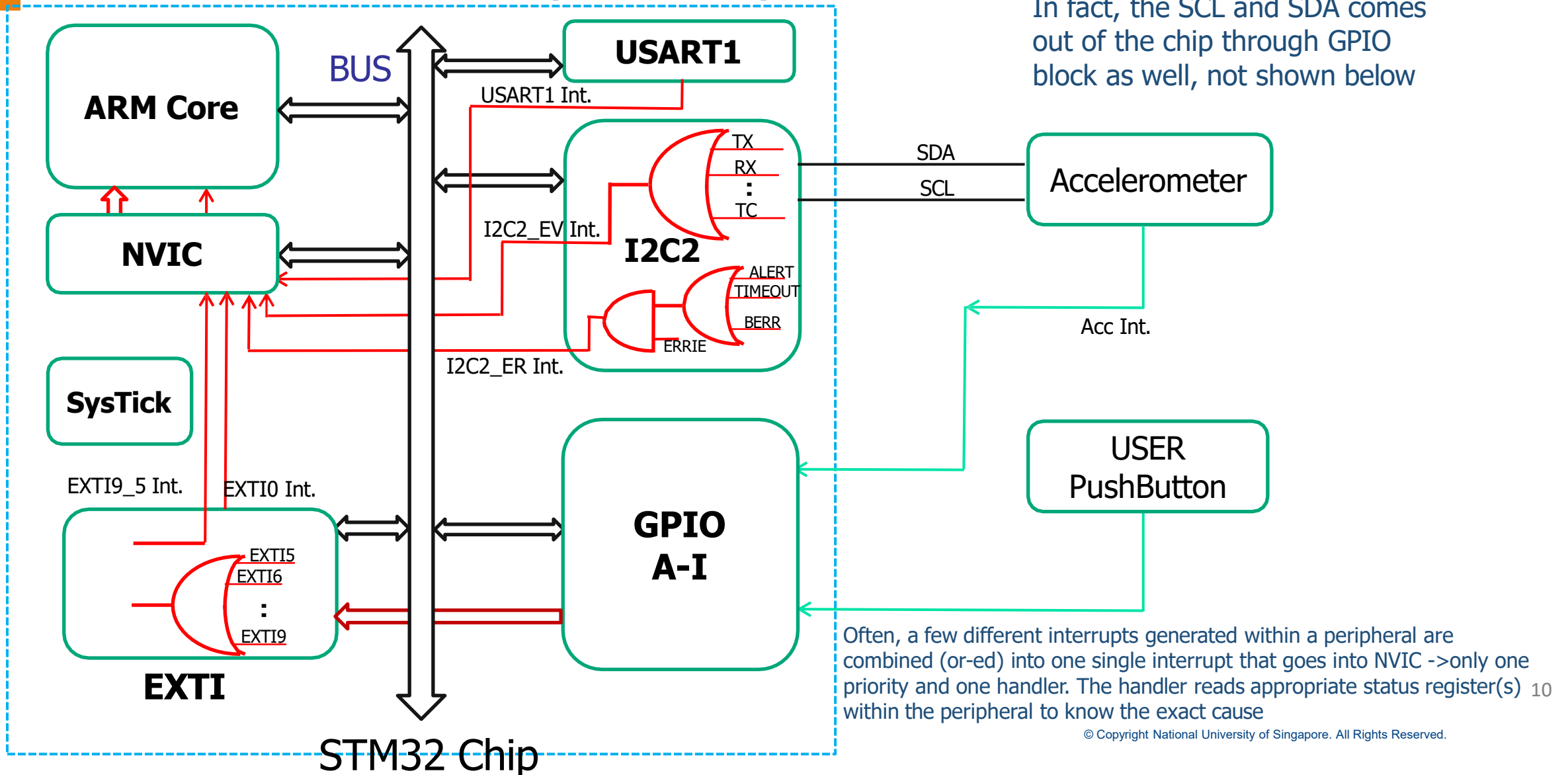
Table 7.1 List of System Exceptions

Exception Number	Exception Type	Priority	Description
1	Reset	-3 (Highest)	Reset
2	NMI	-2	Nonmaskable interrupt (external NMI input)
3	Hard fault	-1	All fault conditions if the corresponding fault handler is not enabled
4	MemManage fault	Programmable	Memory management fault; Memory
5	Bus fa		
6	Usage		
7-10	Reserved	NA	—
11	SVC	Programmable	Supervisor Call
12	Debug monitor	Programmable	Debug monitor (breakpoints, watchpoints, or external debug requests)
13	Reserved	NA	—
14	PendSV	Programmable	Pendable Service Call
15	SYSTICK	Programmable	System Tick Timer

Table 7.6 Exception Vector Table After Power Up

Address	Exception Number	Value (Word Size)
0x00000000	—	MSP initial value
0x00000004	1	Reset vector (program counter initial value)
0x00000008	2	NMI handler starting address
0x0000000C	3	Hard fault handler starting address
...	...	Other handler starting address

NVIC CONNECTIONS (PARTIAL)



CONCLUSION

- Introduction to Cortex-M4 NVIC
- Cortex-M4 Interrupt system
 - Exceptions
 - Interrupts
- Vector Table
- NVIC Connections (Partial)